

ERROR HANDLING IN COMPILERS

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INTRODUCTION

What is Error Handling?

Error handling is a process used by a compiler to detect and report errors in a program.

It helps programmers identify and fix mistakes.

The compiler tries to continue checking the remaining code after finding an error.

Good error handling improves software quality.

It reduces development time by helping programmers fix errors quickly.

Every modern programming language compiler includes an error-handling mechanism.

The main goal of error handling is to detect errors early and help produce error-free programs.

TYPES OF COMPILER ERROR

1. Lexical Error

Invalid characters or identifiers

```
unoin test  
{  
    int x;  
    float y;  
} T1;  
→  
union test  
{  
    int x;  
    float y;  
} T1;
```

2. Syntax Error

Violates the grammar rules of the language.

Example: Missing) or ;

```
1  if (x > 5 {  
2      printf("Hello");  
3  }  
4  
Error: missing ')'
```

3. Semantic Error

Syntax is correct, but the statement has the wrong meaning.

```
1  int age;  
2  age = "Twenty";  
3  printf("Age: %d", age);
```

4. Logical Error

Program runs but gives incorrect output.

Example: Using - instead of +

```
1  int a = 10, b = 5;  
2  int sum = a - b; // Should be a + b  
3  printf("Sum = %d", sum);  
4  
Output  
Sum = 5  
→  
!  
Wrong Result
```

5. Runtime Error

Occurs while the program is running.

Example: Division by zero.

```
1  int x = 10, y = 0;  
2  int z = x / y; // Division by zero  
3  printf("Result = %d", z);  
↓  
Program.exe  
! Runtime Error!  
Division by zero.  
OK
```


ERROR HANDLING TECHNIQUES

1. Panic Mode Recovery

The compiler skips incorrect code until it finds a safe point (such as ; or }).

Simple and commonly used method.

2. Phrase-Level Recovery

The compiler makes small corrections, such as inserting or deleting a missing symbol.

Allows compilation to continue with minimal interruption.

3. Error Productions

The compiler uses predefined grammar rules to detect common programming mistakes.

Provides more specific and helpful error messages.

4. Global Correction

The compiler determines the minimum changes needed to make the program correct.

Produces accurate results but is rarely used because it is time-consuming.



ADVANTAGES AND APPLICATIONS

Advantages

- Finds errors early during compilation.
- Makes debugging easier with clear error messages.
- Improves program quality and reliability.
- Saves time by helping fix errors quickly.

Applications

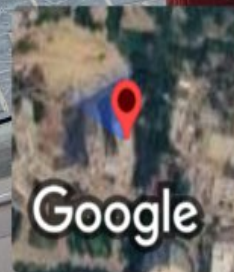
- Used in programming language compilers.
- Used in code editors and IDEs (like Visual Studio Code).
- Used in software development and testing.
- Used in educational coding platforms to help students learn programming.



Thank You



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